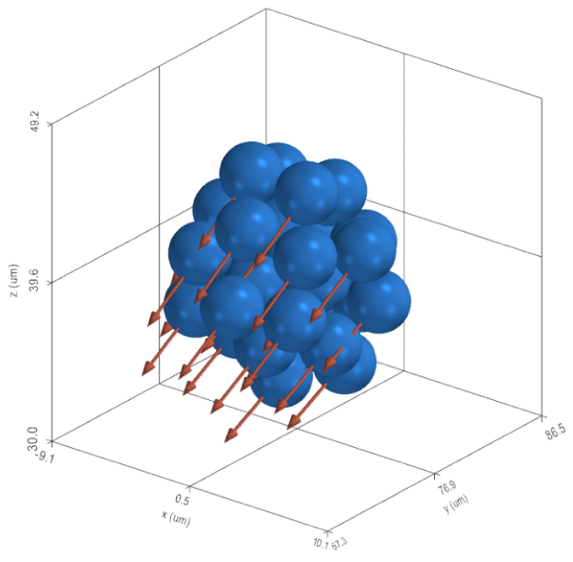
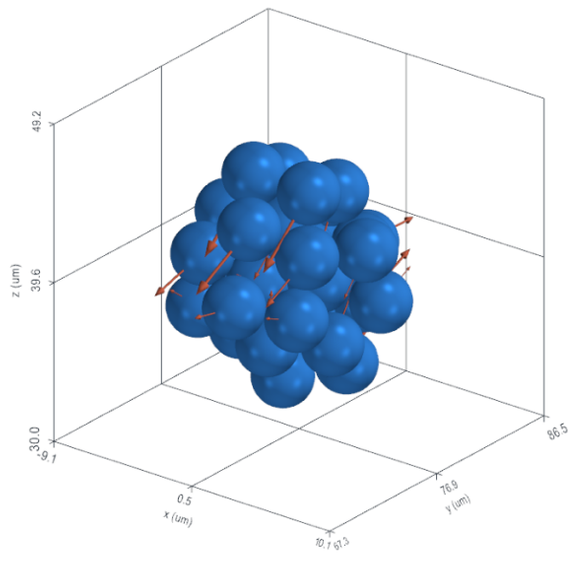
F3-1 | Differential acoustic driving at the same mature state

Actual spheres; arrows alone are scaled · surface-anchored vectors · no particle displacement

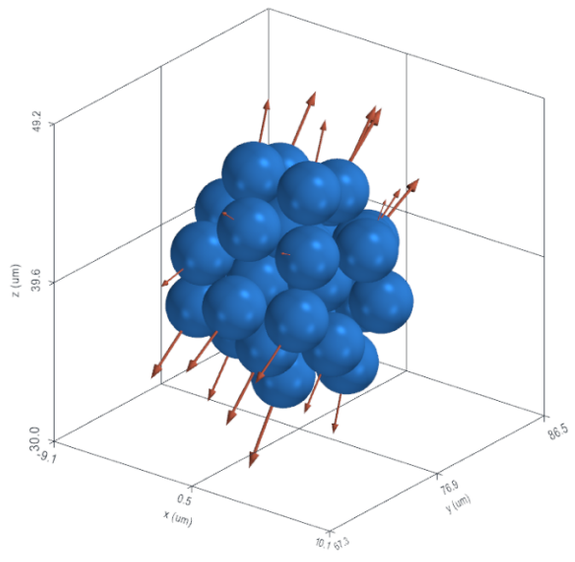
a Unchanged: Primary force
Arrow scale: 1 pN = 20.1 µm



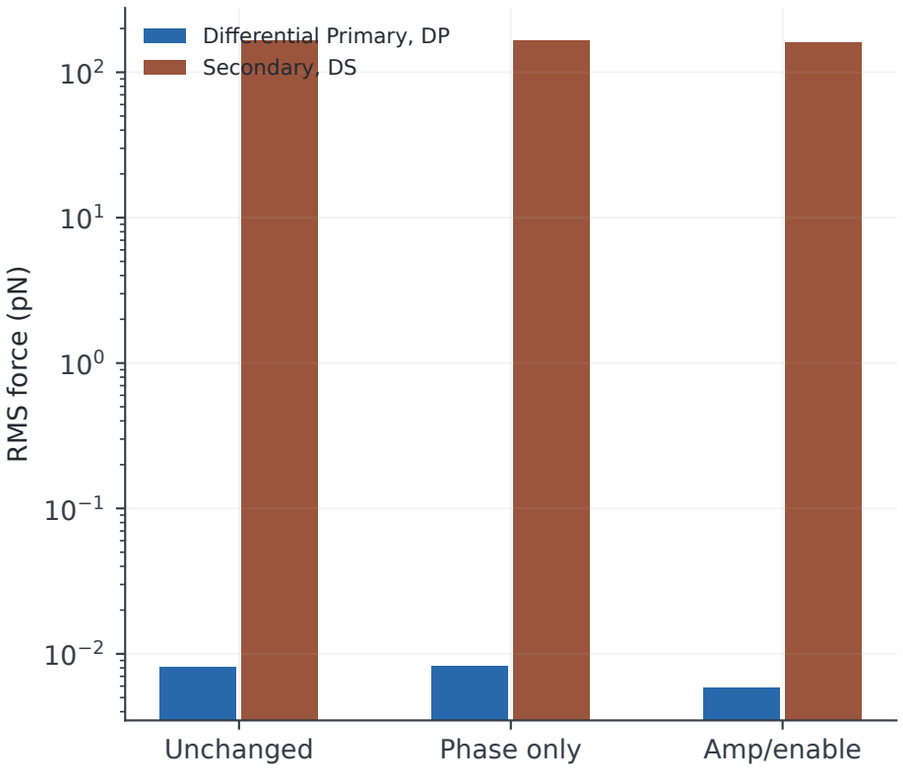
b Phase only: Centered control-induced Primary
Arrow scale: 1 pN = 1038.4 µm



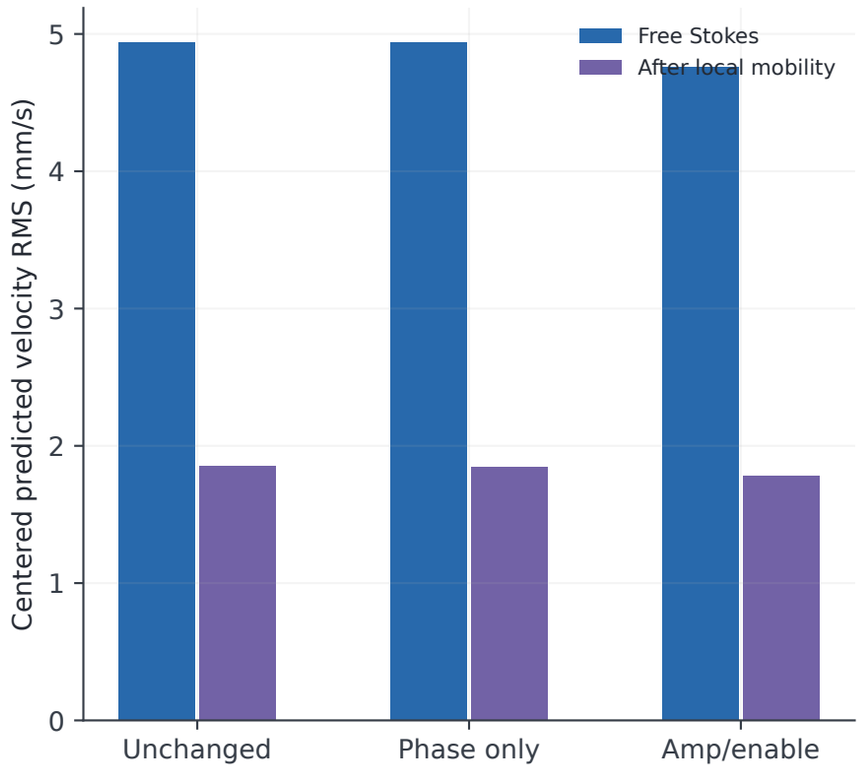
c Amp/enable: Centered control-induced Primary
Arrow scale: 1 pN = 521.7 µm



d Direct differential drive is small



e Pre-contact motion; not observed strain



f Resolution and interpretation

Grid spacing: 100 µm; occupied cells: 2
32 actual positions; R = 1.78 µm

DP / DS ≈ 3.6–5.0 × 10^{-5}
Stored-gradient vs pressure-derivative
L2 mismatch: 24–38% (eligible controls)

Amplitude control: 72.3% of Secondary
force-change energy is global scaling.

Local mobility also leaves a torque-balance
residual; it is not an exact coupled solve.
Frozen physics retained; EXPLORATORY.